Science · LAUNCH · Autumn 2 · Week 2 · SCI-LAUNCH-A2-W02-L3

Guidance and staff card

At check-out, record E for … against the pupil's target.

At check-out, record E for using evidence about growth and differentiation to repair a stem-cell claim against the pupil's target.

Prepared 3 November 2026

Invite a response. Give thinking time. The learner may pass. Offer private answers and the usual response mode.

Return to learning: use the individual pause plan, give space, then return to the last secure step and one structured choice. Correct privately.

Use soft instrumental music for transitions if the class plan says so.

Code	Meaning
VF	Verbal feedback given here
WS	Worked with support
I	Independent
NS + …	Next step one specific thing
E	Evidence captured on the evidence app
R	Responded loop closed after a pupil response and adult engagement
//	Self-edit point
?	Read this back to me

At check-out, record E for today's lesson against each pupil's target on the evidence app

Use observed evidence and record the support given. An edit is in the same colour, on the same page. Written feedback is used for extended writing when it adds value.

Codes follow the verified feedback policy, Issue 1. A button, saved answer or completed stage does not award a code.

Feedback codes VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil’s improvement to the work.
Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt

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Teacher and cover plan

At check-out, record E for using evidence about growth and differentiation to repair a stem-cell claim against the pupil's target.

I am learning to explain growth and differentiation and use evidence to repair claims about stem cells.

Total: 40 minutes. Start from the actual responses. Keep one comparison, a repaired claim, review and the exit.

Week arc: Discover: growth and stem cells. Use: how stem cells become specialised cells. Master: growth, differentiation and stem-cell evidence.

Kit: large prepared process cards, desk mat and labelled references. Build in quiet resets during shared work, independent work and review.

Teaching sequence

Stage	Minutes	Teaching and check
Arrival	5	Q1 checks picture access. Use the stem cell label before asking about division. Offer enlarged or tactile paper shapes; accept pointing before speaking.
Starter	3	Ask the original three retrieval questions in order. Invite a response and allow a pass. Reveal answers separately and revisit the success criteria.
I do	3	Trace the skin model. Ask what changes in number and what changes in role. Use large cards if the drawing is hard to access.
We do	7	Invite each learner to identify the evidence and repair the claim. Start with root length, then switch to the blood-cell lineage. Hear why the repair fits. Use the optional participation prompt and check the effect of any agreed change.
I do	2	Step through the model. A root can lengthen through more cells, longer cells or both. The drawn sequence is simplified.
We do	5	Hear an individual reason for each link. In the fresh case, the model supports an increase in number. Ask what further evidence would help identify a specialised type.
You do	8	Hear a comparison and a repaired claim in the learner’s usual mode. Record actual independence and support.
Review and improve	4	Give one specific feedback point. Hear the edit or explanation and use it to choose the next prompt.
Exit ticket	3	Hear every response through the usual mode. Use it to change the next example or prompt. Keep one response and state the support actually used.

Stage timings

Stage	Minutes
Today I can	0
Arrival	5
Starter	3
I do	3
We do	7
I do	2
We do	5
You do	8
Review and improve	4
Exit ticket	3
Tools & print	0

Total: 40 minutes

Feedback codes VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil's improvement to the work.
Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Exit pupil voice
What helped you most today, or what would you change next time?
Accept a pointed, spoken, signed, drawn, written or chosen response. A pupil may pass.

Made by Matt

P17 · SLT planning and evaluation

Planned adaptations

Need	Planned adaptation
Cognition and learning	Use one process and one visible change, then fade one cue.
Communication and interaction	Rehearse in pairs or respond by pointing, signing or exact scribing.
Regulation	Use the individual pause plan and return to one model choice.
Sensory and physical	Offer enlarged cards and adult-directed handling.
Language access	Pre-teach number, length, role and evidence with a picture.

If behind

Use one case in the repair model. Keep an independent explanation, review and exit. Record any change to coverage.

Where catch-up is embedded

Use the picture at Arrival and the static references beside each model. The catch-up sheet replaces the first step after absence or an access barrier.

Staff evaluation

Prompt	Record after teaching
Delivery	Actual change and reason
Criteria	Response and support used
Adaptation	Change and effect on a fresh response
Coverage	Remaining teaching move
Evidence	Location on the evidence app and lesson code

Made by Matt

P18 · Individual target opportunities

Use each learner’s existing targets. Leave pupil codes and target fields blank.

Opportunity	Lesson evidence and adult response
Cognition and learning	Match a process to an outcome, then fade the cue.
Communication	Give a claim repair using the agreed response mode.
Regulation	Request an adjustment and return to one step. Check the effect.
Sensory or physical access	Direct a choice or place a card with usual equipment. Record access support separately.
Language access	Use one precise science term in a fresh response.

Made by Matt

Adult deployment and responsive feedback

Agree who leads the model and who checks access. Hear the learner’s reason privately if needed. Wait, point, rephrase and then offer a limited choice. Check a fresh case after prompting.

For the optional participation prompt, hear the learner, agree a possible change, explain any limit and check whether it helped.

Made by Matt

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First lesson back · once-only read

Before the lesson · scan the books

Scan for five minutes. Find where teaching stopped and what responses were actually made. Look for access barriers and useful science evidence.

Existing NS + is history

Do not chase, back-mark or retrospectively close earlier feedback. Begin fresh feedback from this lesson.

Ask the cover teacher and TAs

Ask what was taught, which models were used, what changed for individuals and which responses were heard.

Check the roll

Compare attendance with the available evidence. New arrivals start from what they show here; absence creates no automatic deficit label.

Arrival read-off · decide by about minute eight

Q1 checks picture access. Use Q2 for number, Q3 for inherited information and Q4 for plant growth. If number and job are confused, use the first model. If secure, fade the cue and use a new source-target example.

After last lesson · one read, once

Read each exit against its criterion and the support used. Choose the next route from the least secure dependency. Teach next; do not reopen historical feedback.

Made by Matt

Preparation and practical safety

This is a model lesson. It uses prepared cards and drawings. It does not involve culturing cells, collecting tissue, handling blood, treatment decisions or a claimed core practical.

Check large cards and the desk area. Keep loose cards on a mat and pick up fallen items. Provide seated or adult-directed handling where needed. Follow the local risk assessment and individual plans.

Test the offline display, keyboard controls, print choices and reset actions. Keep a static reference available. Use each learner’s usual access equipment.

Destination link: explaining cell structure and interpreting scientific claims supports further biology study and work in laboratory support and biomedical science.

Made by Matt

Teacher answers and topic guidance

Accept equivalent science in the usual response mode. Keep answers hidden during independent attempts.

Starter

1. Cell division.
2. Differentiation.
3. In the root-tip meristem.

Repair an evidence claim

A longer root supports growth. Length alone cannot establish cell identity or the relative contribution of division and elongation. The blood-cell drawing models a possible lineage for the named adult source. It cannot prove a treatment outcome or a wider cell range.

Build the growth chain

Division increases number. Elongation increases length. Differentiation gives a specialised role. For the fresh skin case, more cells support division. Look for features or function before naming a specialised type.

Scope

Embryonic stem cells can give rise to many body-cell types. Adult blood-forming stem cells have a narrower range. Cells in root-tip meristems divide. In skin repair, new cells can replace damaged cells.

Arrival Supported

1. Stem cell.
2. More cells.
3. The nucleus.
4. Root tip.

Arrival Independent

1. Stem cell.
2. More cells with matching inherited information.
3. Division makes more cells; differentiation gives a specialised role.
4. In the root-tip meristem.

Arrival Aspire

1. Stem cell.
2. Inherited DNA is copied; two daughter cells receive matching inherited information.
3. Division is shown. Differentiation explains specialised roles.

4. The new cells can elongate, increasing their length.

Exit Supported

1. Division.

2. The root grew.

Exit Independent

1. Division increases cell number.

2. Evidence of a characteristic feature or function.

Exit Aspire

1. Cell count shows a change in number. Differentiation requires evidence about specialised features or roles.

2. It models a possible blood-cell lineage. Treatment success would require outcome evidence.

Made by Matt

Mark scheme Supported

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Match division, elongation and differentiation to more cells, longer cells and a specialised job. [3]

Division adds cells. Elongation lengthens cells. Differentiation gives cells features suited to a job.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Compare damaged skin and a growing root. Name one shared process and one feature of the root route. [2]

Both use division. In the root, division happens in the meristem or new cells can elongate.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. A root is longer. Choose what the evidence supports: the root grew, or every root cell is a stem cell. Explain your choice. [2]

The root grew. Length alone does not identify the types of cells in it.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Independent

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Explain how division and differentiation can help replace damaged skin cells. [3]

Division adds cells. Some descendants differentiate and acquire features suited to their roles. New cells replace damaged cells.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Compare skin repair and root growth. Use division, meristem and elongation. [3]

Both use division to add cells. Root-tip meristem cells divide. New root cells can elongate, contributing to length.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Repair this claim: more cells in a tissue proves that new specialised cell types formed. Use the cell count and name one extra check. [3]

The count supports an increase in cell number. It does not identify specialised types. Compare features with a labelled reference or gather evidence of their function.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Aspire

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. A root is longer and the model shows longer cells. Explain how division and elongation could contribute. State what length alone cannot establish. [3]

Division can add cells. Elongation increases cell length. Root length alone does not tell which process contributed or identify all cell types.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Repair: an adult blood-forming stem cell can become any body cell. Use the blood-cell example and compare it with an embryonic source. [3]

Adult blood-forming stem cells give rise to blood-cell types. Embryonic stem cells can give rise to many body-cell types. The adult example does not support the claim about any body cell.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. A model labels every new skin cell as a stem cell. Correct two labels or claims and justify your edits with number and role evidence. [4]

Use division for the increase in cell number and justify it using the count. Identify specialised descendants only when there is evidence of features or a role and link this to differentiation.

Maximum 4 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Choice

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Choose division or differentiation for more cells. Choose elongation or division for longer cells. Give one reason. [3]

Division adds cells. Elongation increases length. A reason distinguishes the two changes.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Choose both, skin only or root only for making more cells by division. Name one root feature. [2]

Both. Accept a root-tip meristem or elongation of new root cells.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. A root is longer. Choose growth or proof of stem-cell identity as the supported claim. Give a reason. [2]

Growth. Length does not tell us the identity of each cell.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Science staff guidance

Invite a response, give thinking time and allow a pass. Separate access support from help with the science. Use pointing, sign, drawing, exact scribing or adult-directed handling.

Check number, length and role separately. If a claim goes beyond the observation, ask the learner to name what was measured and one further check.

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TA Brief

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I am learning to explain growth and differentiation and use evidence to repair claims about stem cells.

Division adds cells. Elongation lengthens cells. Differentiation gives cells features suited to a role.

At the repair model, hear what the evidence shows before offering a claim. Root length shows growth. The blood-cell drawing models a possible lineage. Ask the learner for one limit.

Wait, point and rephrase before giving a content cue. Record actual I · S · C beside the task. Use the agreed quiet pause and return to one step.

Feedback codes VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil's improvement to the work.
Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Exit pupil voice
What helped you most today, or what would you change next time?
Accept a pointed, spoken, signed, drawn, written or chosen response. A pupil may pass.

Stage timings

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I do	3
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We do	5
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Review and improve	4
Exit ticket	3
Tools & print	0

Total: 40 minutes

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